

Q1. (Writing Pseudocode) [5 Marks]

Write a pseudocode to read N exam scores from the user and output: (a) the highest score, (b) the number of students who scored above the average, and (c) the lowest score after dropping the first and last input. Assume $N \geq 3$.

#1. Read the total number of scores (Assuming $N \geq 3$)

READ N from user

#Initialize variables to store data and calculate metrics

scores = empty list []

total_sum = 0

highest_score = -1

Assuming scores are positive, or use -infinity

#2. Loop to read all N scores

FOR i FROM 0 TO N - 1:

 READ current_score from user

 APPEND current_score to scores

 total_sum = total_sum + current_score

 # (a) Check and update the highest score

 IF current_score > highest_score:

 highest_score = current_score

#Calculate the average score

average_score = total_sum / N

#3. (b) Find the number of students who scored above the average

above_avg_count = 0

FOR EACH score IN scores:

IF score > average_score:

above_avg_count = above_avg_count + 1

#4. (c) Find the lowest score after dropping the first and last input

Since $N \geq 3$, we look at indices from 1 to $N-2$

lowest_trimmed_score = scores[1]

Initialize with the second element

FOR i FROM 1 TO $N - 2$:

IF scores[i] < lowest_trimmed_score:

lowest_trimmed_score = scores[i]

#5. Output the final results

PRINT "Highest score: ", highest_score

PRINT "Students above average: ", above_avg_count

PRINT "Lowest score (excluding first/last): ", lowest_trimmed_score